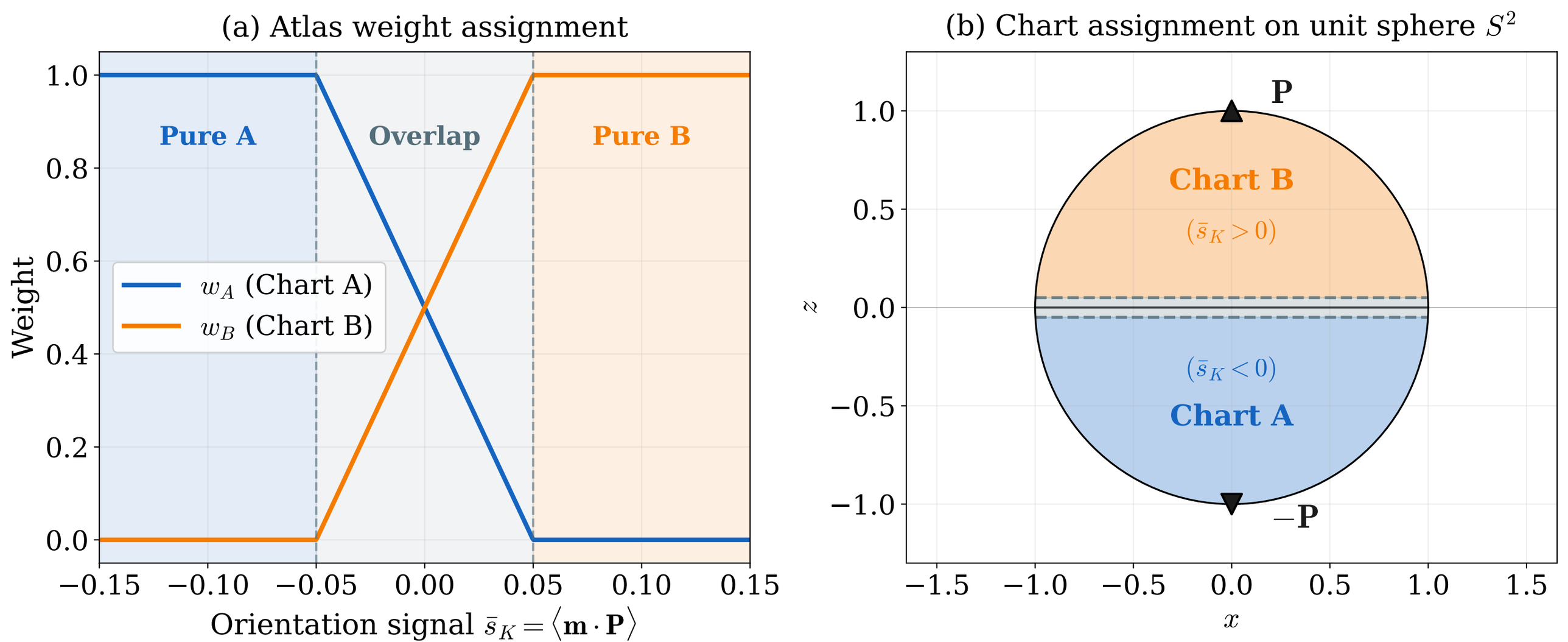
(a) Atlas weight assignment
Pure A
Overlap
Pure B
$w_A$ (Chart A)
$w_B$ (Chart B)
Weight
1.0
0.8
0.6
0.4
0.2
0.0
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Orientation signal $\bar{s}_K = \langle \mathbf{m} \cdot \mathbf{P} \rangle$
(b) Chart assignment on unit sphere $S^2$
P
Chart B
$(\bar{s}_K > 0)$
$(\bar{s}_K < 0)$
Chart A
−P
1.0
0.5
0.0
−0.5
−1.0
$z$
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
$x$